1 Question 1

Question text

(origin) at (0,0); (negx) at (-5,0); (posx) at (5,0); (negy) at (0, -5); (vecb) at (2.4075, -9.1945); (veca) at (-2.4075, 3.1945);
[<->,ultra thick] (-5,0) coordinate (A) –(5,0) node[right]x; [<->,ultra thick] (0,-5)–(0,5) node[above]y;
[->, thick, capri] (0,0) coordinate (B) – (-2.4075, 3.1945) coordinate (C) node[left]**A**;
pic[thick, capri, "53°", draw, <-,angle radius=1cm,angle eccentricity=1.4] angle = veca–origin–negx;
pic[thick, "127°", draw, ->,angle radius=1cm, angle eccentricity=1.4] angle = posx–origin–veca;
[thick, capri, decorate,decoration=brace,amplitude=6pt,mirror,raise=1ex] (0,0) – (-2.4075, 3.1945) node[midway, yshift=2em, xshift=2em]8.0 m;
[->, thick, carminered] (0,0) coordinate (B) – (2.4075, -9.1945) coordinate (C) node[left]**B**;
pic[thick, carminered, "14.7°", draw, ->,angle radius=5cm, angle eccentricity=1.1] angle = negy–origin–vecb;
[thick, carminered, decorate,decoration=brace,amplitude=6pt,raise=1ex] (0,0) – (2.4075, -9.1945) node[midway, xshift=2.5em, yshift=0.5em]19.0 m;
[->, thick, brightlavender] (0,0) coordinate (B) – (0, -6) coordinate (C) node[below]**A + B**;
[thick, brightlavender, decorate,decoration=brace,amplitude=6pt,raise=1ex, mirror] (0,0) – (0, -6) node[midway, xshift=-3em]12.0 m;